UNIT 5 :

FOOD SAFETY AND ETHICS

Q: Discuss in detail about Food safety and ethics

A:

Once food is harvested, enzymes and bacteria cause it to decay and spoil. Food safety involves proper handling, storage, and preservation to prevent contamination and spoilage. Safe food is free from harmful contaminants, toxins, and microorganisms that could cause health issues.To ensure that food is safe for consumption, it should be:

- Protected from contamination by harmful bacteria, poison and other foreign bodies
- Prevented from having any bacteria present multiplying to an extent which would result in the illness of consumers or the early spoilage of the food
- For some foods, thoroughly cooked to destroy any harmful bacteria present
- Discarded when spoilt and/or contaminated

The benefits of proper food safety and hygiene are:

- More efficient utilization of food consumed by the body contributing to improved health and nutrition outcomes
- Prevention of food-borne illnesses (and sometimes death)
- Less food wastage

Causes of food contamination : Food contamination can occur due to poor handling, hygiene, or food adulteration, while spoilage is caused by enzymes or microorganisms like bacteria, yeasts, and moulds. Bacteria multiply rapidly in warm, moist conditions, but can be controlled by boiling or slowed by cold temperatures.

Identifying spoiled food: Spoilt food can have off-odors, discolouration, slime, gas production, unusual taste, or mould growth.

Foods at high risk of food spoilage: Protein-rich and ready-to-eat foods like meat, milk, and eggs spoil faster.

Food cross-contamination: Safe food can spoil when contaminated by bacteria from other foods or surfaces.

Avoiding cross-contamination: Keep raw and cooked foods separate, use clean utensils, and maintain hygiene.

General tips for preventing food contamination and spoilage:

- Food hygiene
- Personal hygiene
- Kitchen hygiene
- Environmental hygiene

Food Additives

Q: Classify food additives? Discuss the limits of additives to be added to the food as per the standards.

A:

Classification of Food Additives

Food additives are substances added to food to enhance its flavor, appearance, shelf life, or other qualities. They are broadly classified into different categories based on their function:

1. **Preservatives:** These prevent spoilage caused by bacteria, fungi, or other microorganisms, extending the shelf life of food. Examples: Sodium benzoate, Potassium sorbate, Nitrites
2. **Antioxidants:** These prevent the oxidation of fats and oils, which can cause rancidity. Examples: Ascorbic acid (Vitamin C), Tocopherols (Vitamin E), Butylated hydroxyanisole (BHA)
3. **Coloring Agents:** These enhance or restore the color of food to make it more appealing. Examples: Tartrazine (yellow), Allura Red, Annatto
4. **Flavor Enhancers:** These improve the existing flavors of food without adding a distinct flavor of their own. Examples: Monosodium glutamate (MSG), Disodium inosinate
5. **Sweeteners:** These add sweetness to food without the calories of sugar. Examples: Aspartame, Sucralose, Stevia
6. **Emulsifiers, Stabilizers, and Thickeners:** These help in maintaining the texture and consistency of food. Emulsifiers help mix ingredients that normally wouldn't blend. Stabilizers and thickeners give food its desired texture. Examples: Lecithin, Pectin, Xanthan gum
7. **Acidulants and pH Regulators:** These maintain or alter the pH levels in food to enhance flavor or stability. Examples: Citric acid, Phosphoric acid
8. **Anti-caking Agents:** These prevent powdered or granulated food products from clumping together. Examples: Calcium silicate, Silicon dioxide
9. **Humectants:** These help retain moisture in food products, keeping them fresh. Examples: Glycerol, Sorbitol
10. **Leavening Agents:** These release gases that cause dough or batter to rise during baking. Examples: Baking powder, Sodium bicarbonate

Limits of Additives as per Standards

The use of food additives is regulated by food safety authorities like the Food Safety and

Standards Authority of India (FSSAI), the Food and Drug Administration (FDA), and the European Food Safety Authority (EFSA). Additives must be used within safe limits to avoid potential health risks. These limits are typically determined based on the Acceptable Daily Intake (ADI), which is the maximum amount of an additive that can be safely consumed daily over a lifetime without health risks.

Key Guidelines for Limits of Additives:

- **Maximum Permissible Levels:** Food additives must be used at the minimum level necessary to achieve their purpose. For example, preservatives like sodium benzoate in beverages are typically limited to 0.1% by weight.
- **Good Manufacturing Practices (GMP):** Many additives are permitted under GMP, meaning they can only be added in amounts that are sufficient to achieve the desired effect, and must not exceed safety limits.
- **Categorical Limits:** Different food categories may have different limits for the same additive. For instance, coloring agents may have a higher permissible level in candy than in beverages.
- **Additive Combinations:** The use of combinations of additives (such as preservatives and antioxidants) is also regulated to prevent excessive intake of a single additive. If multiple additives are used in a food product, their combined total must not exceed the specified safe limit.
- **Prohibited Additives:** Some additives are entirely banned due to their potential harmful effects. For example, the use of Cyclamate as a sweetener is banned in many countries.
- **Labeling and Transparency:** All additives used in food must be declared on the label, along with their specific function. This allows consumers to make informed choices and also helps in monitoring the intake of these additives

Food Allergen Drugs

Q: Discuss the details about food and drug administration.

Antibiotics in Animals

Antibiotics are naturally-occurring, semi-synthetic and synthetic compounds with antimicrobial activity that can be administered orally, parenterally or topically and are used in human and veterinary medicine to treat and prevent disease, and for growth promotion in food animals.

Moreover, antibiotics are also used to improve performance in growth and feed efficiency, to synchronize or control the reproductive cycle and breeding performance also often lead to harmful residual effects.

The use of antibiotics in animals can have direct and indirect effects on human health. The direct effects are those that can be causally linked to contact with antibiotic-resistant bacteria

from food animals, and indirect effects are those that result from contact with resistant organisms that have been spread to various components of the ecosystem (e.g., water and soil) as a result of antibiotic use in food animals.
The concern over antibiotic residues in food of animal origin occurs in two situations; one which produces potential threat to direct toxicity in human, second is whether the low levels of antibiotic exposure would result in alteration of microflora, cause disease and the possible development of resistant strains which cause failure of antibiotic therapy .

The use of antibiotics in food animals

- **Antibiotics as growth promoters:** The mechanism of action of antibiotics as growth promoters is related to interactions between the antibiotics and the gut microbiota. The low doses of antibiotics are sometimes added to cattle, poultry and swine feed to increase their body size. Example includes Sulphonamides which is used as growth promoter in poultry.
- **Antibiotics as therapy,and prophylaxis:** The use of antibiotics in specific conditions is justified because the role of microbial agents is mainly to kill or destroy the rapidly invading cells. These invading microorganisms sometimes damage the cells of the animals and reduce their growth performance. These antibiotics may either be administered to prevent disease or during an infection.
- **Antibiotic residues:** Antibiotic residues are metabolites found in trace amounts in any edible portion of the animal product after the administration of the antibiotics. The antibiotic residues in food animal in excess of the acceptable maximum residue limit may contribute to the development of antibiotic resistances in animals or humans.
- **Health impacts:** Antibiotics used in food animals can cause health hazards due to their secretion in edible animal tissues in trace amounts. Some drugs have the potential to produce toxic reactions in consumers directly while some other is able to produce allergic or hypersensitivity reactions For example, b-lactam antibiotics can cause cutaneous eruptions, dermatitis, gastro-intestinal symptoms and anaphylaxis at very low doses.

Such drugs include the penicillin and cephalosporin groups of antibiotics. These direct effects may include the induction of resistance in normal flora of the human gastrointestinal tract due to the consumption of antibiotic-containing meat products causing an outbreak of resistant diarrheal disease. Moreover, increased risk of resistant colonization or infection in humans due to their exposure to farm animals treated with antibiotics.

The Indirect and long term hazards include microbiological effects, carcinogenicity, reproductive effects and teratogenicity. Microbiological effects are one of the major health hazards in human beings. The Resistant bacteria from animal waste used as fertilizer may cause contamination of water supply and alterations in human flora

The risk factors that contribute to foodborne illness

These are the top five risk factors contributing to foodborne illness:

- Food from unsafe sources
- Inadequate cooking
- Improper hot/cold holding temperatures
- Contaminated equipment
- Poor personal hygiene

Contributing factors are preventable causes of outbreaks.

- Contributing factors are the practices and factors that lead to outbreaks. In an outbreak, contributing factors are the “how” and root causes are the “why.”
- Identifying contributing factors in foodborne illness outbreaks can help stop them and prevent them in the future.

Contributing factors fall into 3 types.

- Contamination: Pathogens and other hazards get into food.For example, a sick food worker handles food with their bare hands.
- Proliferation: Pathogens already in food grow. For example, food held at the wrong temperature for a long time can allow harmful bacteria to grow.
- Survival: Pathogens survive a process to kill or reduce them. For example, food is not cooked long enough or to a hot enough temperature.

The most common contributing factors for outbreaks in restaurants come from sick food workers and food preparation practices.

Sick food workers can contaminate ready-to-eat food when they come into work while experiencing symptoms, such as vomiting or diarrhea. Food workers who are infectious can contaminate food:

- with bare hands
- while wearing gloves
- in other ways, such as letting food touch a contaminated cutting board or utensil

Improper food preparation practices are another common contributing factor for restaurant outbreaks. Not cooking food to a hot enough temperature and other improper food preparation practices can lead to pathogens growing.

Major foodborne illnesses and causes

Foodborne illnesses are usually infectious or toxic in nature and caused by bacteria, viruses, parasites or chemical substances entering the body through contaminated food. Chemical contamination can lead to acute poisoning or long-term diseases, such as cancer. Many foodborne diseases may lead to long-lasting disability and death. Some examples of food hazards are listed below.

Bacteria

Salmonella and Escherichia coli are some of the most common foodborne pathogens that affect millions of people annually, sometimes with severe and fatal outcomes. Symptoms can be fever, headache, nausea, vomiting, abdominal pain and diarrhoea. Foods involved in outbreaks of salmonellosis include eggs, poultry and other products of animal origin. Foodborne cases with Campylobacter are mainly caused by raw milk, raw or undercooked poultry and drinking water. E-coli is associated with unpasteurized milk, undercooked meat and contaminated fresh fruits and vegetables.

Listeria infections can lead to miscarriage in pregnant women or death of newborn babies. Although disease occurrence is relatively low, Listeria's severe and sometimes fatal health consequences, particularly among infants, children and the elderly, count them among the most serious foodborne infections. Listeria is found in unpasteurised dairy products and various ready-to-eat foods and can grow at refrigeration temperatures.

Vibrio cholerae can infect people through contaminated water or food. Symptoms may include abdominal pain, vomiting and profuse watery diarrhoea, which quickly lead to severe dehydration and possibly death. Rice, vegetables, millet gruel and various types of seafood have been implicated in cholera outbreaks.

Antimicrobials, such as antibiotics, are essential to treat infections caused by bacteria, including foodborne pathogens. However, their overuse and misuse in veterinary and human medicine has been linked to the emergence and spread of resistant bacteria, rendering the treatment of infectious diseases ineffective in animals and humans.

Viruses

Some viruses can be transmitted by food consumption. Norovirus is a common cause of foodborne infections that is characterized by nausea, explosive vomiting, watery diarrhoea and abdominal pain. Hepatitis A virus can also be transmitted by food and can cause long-lasting liver disease and spreads typically through raw or undercooked seafood or contaminated raw produce.

Parasites

Some parasites, such as fish-borne trematodes, are only transmitted through food. Others, for example tapeworms like Echinococcus spp, or Taenia spp, may infect people through food or direct contact with animals. Other parasites, such as Ascaris, Cryptosporidium, Entamoeba histolytica or Giardia, enter the food chain via water or soil and can contaminate fresh produce.

Prions

Prions, infectious agents composed of protein, are unique in that they are associated with specific forms of neurodegenerative disease. Bovine spongiform encephalopathy (BSE, or so-called mad cow disease) is a prion disease in cattle, associated with the variant

Creutzfeldt-Jakob disease (vCJD) in humans. Consuming meat products containing specified risk material, such as brain tissue, is the most likely route of transmission of the prion agent to humans.

Chemicals

Naturally occurring toxins include mycotoxins, marine biotoxins, cyanogenic glycosides and toxins occurring in poisonous mushrooms. Staple foods like corn or cereals can contain high levels of mycotoxins, such as aflatoxin and ochratoxin, produced by mould on grain. A long-term exposure can affect the immune system and normal development, or cause cancer.

Persistent organic pollutants (POPs) are compounds that accumulate in the environment and human body. Known examples are dioxins and polychlorinated biphenyls (PCBs), which are unwanted by-products of industrial processes and waste incineration. Dioxins are highly toxic and can cause reproductive and developmental problems, damage the immune system, interfere with hormones and cause cancer.

Heavy metals such as lead, cadmium and mercury cause neurological and kidney damage. Contamination by heavy metal in food occurs mainly through pollution of water and soil.

Other chemical hazards in food can include radioactive nucleotides that can be discharged into the environment from industries and from civil or military nuclear operations, food allergens, residues of drugs and other contaminants incorporated in the food during the process.

Consumers Lifestyles and Demands

Consumers want to eat healthily, but the price is an obstacle. The pandemic has put health at the forefront and changed our relationship with food. Consumers want to eat healthier, and this has been the most significant lifestyle change in recent years, followed by working out more often.

In consumer marketing, lifestyle is considered a psychological variable known to influence the buyer decision process for consumers. Lifestyle can be broadly defined as the way a person lives. In sociology, a lifestyle typically reflects an individual's attitudes, values, or world view.

Consumer behavior can be defined as 'those activities directly involved in obtaining, consuming and disposing of products and services, including the decision processes that precede and follow these actions

Understanding consumer behavior

Consumer behavior can be defined as 'those activities directly involved in obtaining, consuming and disposing of products and services, including the decision processes that precede and follow these actions'. There are environmental influences affecting this

behavior such as ethnicity and culture, social group, regional preferences, as well as food availability and household technology. There are also differences among individuals, not only their age and sex, their education, their standard of living, but also their physiological and psychological make-up. Individuals have their own food choice, which to a greater or lesser extent overrides preferences defined by culture or religion. In the last 30 years, multidisciplinary social science research has increased knowledge of food consumer behavior and food choice.

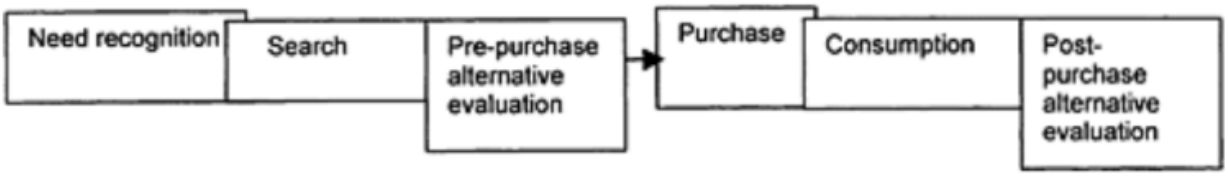


Fig. 5.1 General consumer behaviour in buying and consumption (Source: After Engel et al., 1995).

1. Exposure to information, communication, and the product.
2. Attention given to the information.
3. Comprehension of the information, as it is analyzed against the knowledge and the attitudes stored in the memory.
4. Acceptance or rejection of the incoming information.
5. Retention of the new information in the memory as knowledge.

When buying food, the consumer usually wants to keep the decision process as simple and quick as possible. Food is consumed two or three times a day, and may be bought every day, so the consumer does not want to spend a great deal of time in buying and today even in consuming food. It is only the special occasion, the special meal or the special food that is given detailed analysis. But it is important for the food designer to recognise that there is detailed and critical thought at certain points of time. Consumers receive a great deal of information on food through the media and advertising as well as by word-of-mouth; gradually and often imperceptibly this information changes their knowledge base and therefore their food behavior. If there is a great deal of information on saturated fats in the diet, they will consider and may gradually change to low-fat foods. Sometimes their behavior is changed by a jolt; this could be a food poisoning scare, or it could be food poisoning affecting them directly.

These can cause long-term changes in food behavior, for example reduced meat-eating triggered by reports of BSE ('mad cow disease'), and complete rejection of shellfish caused by a bout of shellfish-related food poisoning.

In the past ten years, food brands in basic food areas have had very similar features and competed mainly on price, so that they have degenerated into commodity products - no brand being distinctive. Also there has been a proliferation of products with very little difference between them - for example in the small New Zealand market, there are about 157 breakfast cereals under the national and retailer brands. Can the consumers differentiate adequately between these, even if they read a consumer magazine? No wonder they choose the easy way out and buy on price, choosing the specials; or keep on buying their familiar product. Foods can be bought on impulse to relieve food boredom or as a treat. A new snack or a new takeaway

can be bought to see if it lives up to its promotion; if it satisfies the consumer it can become a regular food.

The food designer needs to be aware, in the target market, of the general consumer behavior towards foods and eating, and how this is slowly changing with time, but also needs to recognise the sudden change. This can be caused by either new information giving an attitude change, or new foods giving the consumers some greater advantage for safety, nutrition, convenience or attractiveness. Companies that have a long-term relationship with their target consumers build up knowledge about the trends in changes of their behavior, which is invaluable in product development.

Food consumer behavior can include the growing of the food but usually in the urban environment it concerns obtaining food from the supermarket, restaurant or takeaway. In the future it may be more distant with food being ordered through the Internet. Food companies must realize that although their food behavior often starts with the selection of food at the retailer, consumers have an interest in how food is produced and they have their concerns about production and processing.

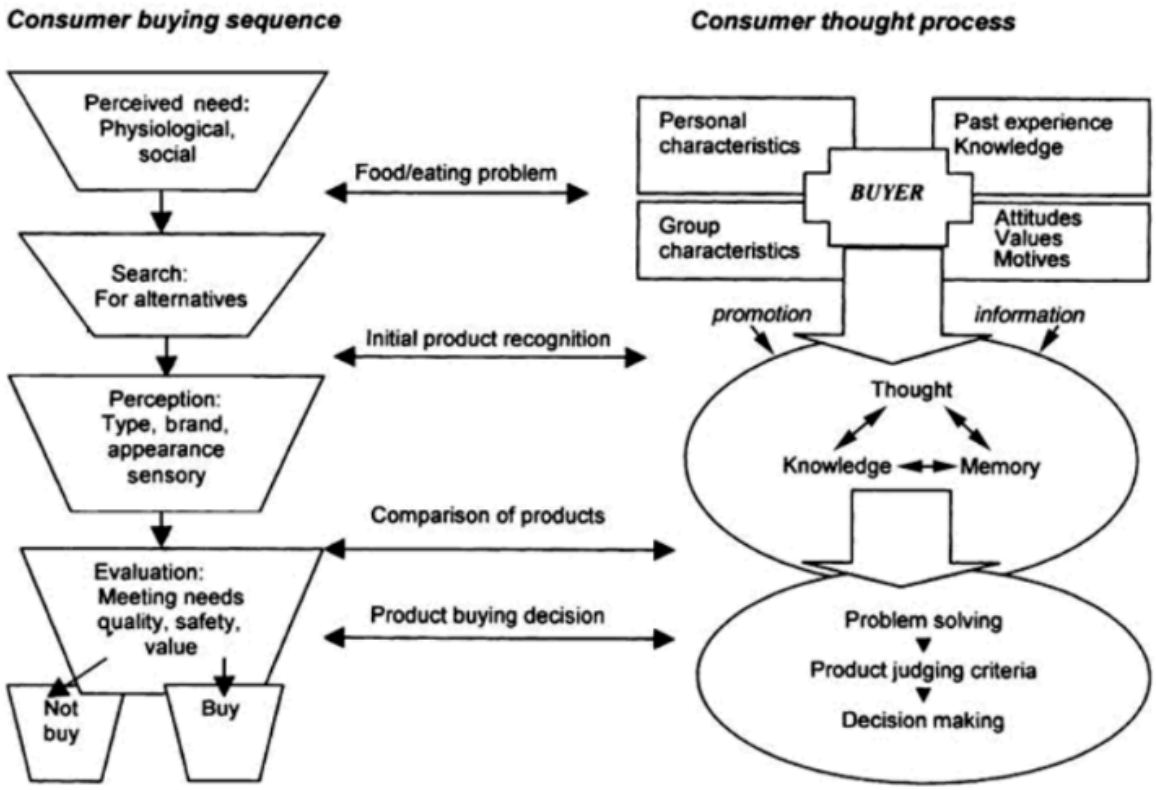


Fig. 5.2 Consumer food purchase patterns.
(- *click to enlarge*)

Recognition of the stimuli that start the buying sequence, the product judging criteria that are used in comparing products during the buying decision, and the level of consumer involvement

with the product throughout the purchase and post-purchase sequence, are important in product design.

Food Production and Economics

Q: Write explanatory notes on Food production and economics

A:

Food production is all about preparing food, in which raw materials are converted into ready-made food products for human use either in the home or in the food processing industries. Its process comprises scientific approaches. Food production has many sections and it starts with basic things like cleaning, packing, segregating, sorting, preparing, adding ingredients in correct proportions, presenting, etc.

How is Food Produced?

There are large numbers of plant and animal products, which are used for our well-being. They provide us with food, which comes from both plants and animals. These include grains, pulses, spices, honey, nuts, cereals, milk, vegetables, fruits, egg, meat, chicken, etc. The existence of our life is completely dependent on plants and animals. Altogether, plant and animal species provide 90% of global energy.

Types of Food Production

cultivation, selection, crop management, harvesting, crop production, preserving, baking, pasteurizing, pudding, carving, butchering, fermenting, pickling, drink and candy making, restaurants, etc.

Methods of Food Production

- Chopping or slicing vegetables
- Curing food.
- Grinding and marinating.
- Emulsification.
- Food fermentation.
- Fermenting beer at brewing industries.
- Boiling, broiling, frying, grilling, steaming and mixing.
- Pasteurization.
- Fruit juice processing.
- Removing the outer layers either by peeling and skinning.
- Gasification of soft drinks.
- Preserving and packaging of food products by vacuum packs.

Food economy : The Food Economy examines how trends like scarcity vs. abundance and slow vs. fast food are interconnected, with various actors shaping the sustainability of food

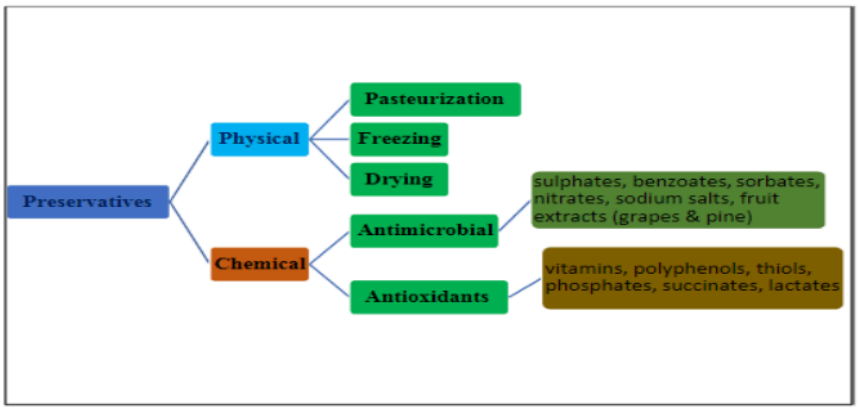
systems. It also highlights challenges to efficient and inclusive food production, distribution, and consumption. A normal good sees increased demand as consumer income rises, with examples like food, clothing, and appliances

Food Economics Division (FED): FED conducts research on food sector policies, consumer food choices, and food assistance programs, while providing data on food prices, expenditures, and supply chains.

- **Food Prices and Markets:** FED monitors food consumption, prices, marketing costs, and price spreads, studying market trends, government interventions, and their impacts on consumers and producers.
- **Food Safety:** FED analyzes food safety, including the economic impact of foodborne diseases, regulatory effects, and consumer responses to food safety events.
- **Food Security and Food Assistance:** FED evaluates SNAP, WIC, and the National School Lunch Program, focusing on food security, program participation, and links to the agricultural sector.
- **Consumer Behavior, Food Consumption, and Nutrition:** FED examines factors influencing food consumption, food availability, consumer preferences for quality and nutrition, and the role of information in shaping food choices.

Role of Preservatives in Food Safety

Preservatives are added to foods to help prevent or slow spoilage due to bacteria, molds, fungi or yeast, as well as to slow or prevent changes in the food's color, flavor or texture, delay rancidity, and maintain freshness.



Physical Food Preservatives:

- **Chilling and freezing:** Chilling and freezing are one of the best physical ways to preserve food.
- **Canning:** It is another way to preserve food by preventing spoilage caused by bacteria and other microorganisms that are present in the surrounding environment. After being canned in boiling water (212°F), jars are cooled quickly

to prevent them from cracking or exploding during processing.
    - **Use high-quality ingredients to preserve food:** The quality of canned products depends on what goes into them.
    - **Dehydrating:** Another way to preserve food is by dehydrating it (i.e., drying); this process uses heat, so one needs to know how much time it takes for the food to be completely dehydrated before starting.
    - **Adding salt:** You may also ferment foods or add salt to allow them to last longer without rotting or use sugar syrup as an addition for pickling vegetables.

Chemical Food Preservatives:

Food preservatives are used to prevent the growth of microorganisms (e.g., bacteria, moulds, and yeasts) that can cause food spoilage. The most common food preservatives are antioxidants and antimicrobials (e.g., BHA, BHT).

Antioxidants prevent the oxidation of fats (lipids) in foods by preventing oxygen from reacting with them.

Antimicrobial agents stop bacterial or mold growth in food by killing microorganisms or inhibiting their growth. The most commonly used antimicrobial agents are benzoates (sodium benzoate and potassium benzoate).

- **Calcium Propionate Powder:** also called calcium salt of propionic acid, is a type of acid, which is a chemical compound that has an acidic taste and odor. Calcium propionate has been used in foods for many years to prevent spoilage and keep foods from becoming discolored or losing flavor or texture.
- **Potassium Sorbate:** an antioxidant that helps prevent spoilage caused by bacteria. It is also used by the food industry as it helps prevent mould growth on fresh fruits and vegetables that have been exposed to the air. Potassium sorbate is often used by businesses in jam and jelly making as a preservative to prevent fruit from turning brown while it sits on the countertop. It's also added to packaged baked goods like cookies, crackers, and pieces of bread to help keep them fresh longer without sacrificing flavour or texture.

The most common chemical preservative used by businesses are:
- Benzoates (like sodium benzoate)
- Nitrites (like sodium nitrite)
- Sulphites (like sulphur dioxide)

Chemical Food Preservatives:

Natural preservatives are an effective way to preserve food. They are a natural, safe way to extend the shelf life of many foods. Natural preservatives for foods can be used in foods and drinks to help prevent spoilage and maintain freshness without artificial ingredients or chemicals.

- **Vinegar:** Vinegar is one of the oldest and safest methods used to preserve foods. It's great for pickling and canning fruits and vegetables, but it also works as a natural preservative in baked goods, condiments, and even salad dressings.
- **Cider Vinegar:** Cider vinegar is made from fermented apples (obviously). It has many health benefits including lowering cholesterol levels, fighting cancer cells, and even helping you lose weight! Cider vinegar also has antibacterial properties thanks to its high acetic acid content (which is formed during fermentation).
- **Citric Acid:** Citric acid is a natural preservative that inhibits spoilage by forming a protective coating on food surfaces which prevents them from absorbing moisture and air, thereby preventing microbial growth and food deterioration.
- **Dill Seeds:** Dill seeds have been used as a preservative since ancient times; they contain an antioxidant called phenolic compounds which are antibiotic and antifungal agents that inhibit bacteria growth on food surfaces such as meats and fish by inhibiting their ability to attach to surfaces or produce colonies

ETHICS

Q: Write explanatory notes on Bioethics and Clinical Ethics

A:

Bioethics : Bioethics is a broad interdisciplinary field that addresses the ethical issues surrounding biological sciences, medicine, and healthcare. It involves the study of moral values and principles as they apply to human life, especially concerning medical practice and scientific research.

Key Aspects of Bioethics:

- **Moral Principles:** In bioethics, the core principles typically involve autonomy, beneficence, non-maleficence, and justice. These principles guide decision-making in healthcare and research.
- **Scope:** Bioethics spans a wide range of topics, including medical ethics, environmental ethics, and research ethics. It covers issues such as human rights, patient consent, the beginning and end of life (abortion, euthanasia), genetic engineering, and organ transplantation.
- **Interdisciplinary Nature:** Bioethics draws from various fields such as philosophy, law, sociology, and theology to analyze ethical issues comprehensively.

Clinical Ethics : Clinical ethics focuses specifically on the ethical dilemmas that arise in the practice of medicine, especially during patient care. It is concerned with applying ethical principles to specific clinical situations, ensuring that decisions are in the best interest of the patient.

Key Aspects of Clinical Ethics:

- **Practical Application:** Clinical ethics is directly related to patient care, ensuring that medical professionals provide care that respects the rights and dignity of patients.
- **Case-based Decision Making:** Clinical ethics often involves case-by-case evaluations of ethical dilemmas, using principles like autonomy (respecting the patient's wishes), beneficence (acting in the patient's best interest), and non-maleficence (avoiding harm).
- **Patient-Physician Relationship:** Clinical ethics emphasizes the importance of communication between the patient and physician, informed consent, and shared decision-making to ensure ethical treatment plans.
- **Confidentiality and Truth-telling:** Issues such as maintaining patient confidentiality and truth-telling are central to clinical ethics. Physicians must balance the need for transparency with the obligation to respect the patient's privacy.

In summary, bioethics covers broader ethical concerns in life sciences and healthcare, while clinical ethics is more narrowly focused on the moral issues encountered in everyday medical practice, particularly in interactions between healthcare providers and patients.

Clinical Ethics

Clinical ethics is a practical discipline that provides a structured approach to assist health professionals in identifying, analyzing and resolving ethical issues that arise in clinical practice.

7 principles of medical ethics:

This approach – focusing on the application of seven mid-level principles to cases (non-maleficence, beneficence, health maximization, efficiency, respect for autonomy, justice, proportionality)

Application of principles of ethics in patient care:

Beneficence, nonmaleficence	Clinical assessment Nature of illness (acute, chronic, reversible, terminal)? Goals of treatment? Treatment options and probability of success for each option? Adverse effects of treatment and does benefit outweigh harm? Effects of no medical/surgical treatment? If treated, plans for limiting treatment? Stopping treatment?
Respect for autonomy	Patient rights and preferences Information given to patient on benefits and risks of treatment? Patient understood the information and gave consent? Patent mentally competent? If competent, what are his/her preferences? If patient mentally incompetent, are patient's prior preferences

	known? If preferences unknown, who is the appropriate surrogate?
Beneficence, nonmaleficence, respect for autonomy	Quality of life (QOL) Expected QOL with and without treatment? Deficits − physical, mental, social − may have after treatment? Judging QOL of patient who cannot express himself/herself? Who is the judge? Recognition of possible physician bias in judging QOL? Rationale to forgo life-sustaining treatment(s)?
Distributive justice	External forces and context Conflicts of interests − does physician benefit financially, professionally by ordering tests, prescribing medications, seeking consultations? Research or educational considerations that affect clinical decisions, physician orders? Conflicts of interests based on religious beliefs? Legal issues? Conflicts of interests between organizations (clinics, hospitals), 3rd party payers? Public health and safety issues? Problems in allocation of scarce resources?

Physicians obligations

- Cure of disease when possible
- Maintenance or improvement of functional status and quality of life (relief of symptoms and suffering)
- Promotion of health and prevention of disease
- Prevention of untimely death
- Education and counseling of patients (condition and prognosis)
- Avoidance of harm to the patient in the course of care
- Providing relief and support near time of death (end-of-life care)

Health policy

National Health Policy is an initiative by the Central Government to strengthen the health system in India. This initiative molds various dimensions of health sectors like disease prevention, promotion of good health via cross-sectoral actions, health investment, strengthening human resources, technological advancements and more.
Launched in 2017, the National Health Policy replaces the existing policy previously established

in 2002. There are significant changes brought to the policy framework and its objectives. National Health Policy was launched in 2017 by the Central Government to replace the existing health policy.

This policy has introduced four significant goals:

- **Changing health priorities:** aims to tackle the increasing non-communicable and infectious diseases in India.
- **Growth of the healthcare industry:** plans to strengthen the healthcare industry by introducing newer and more advanced technologies.
- **Lower the expenditure:** aims to reduce medical expenses and other health-related costs. They aim to provide superior services to poor and backward communities.
- **Economic growth:** aims to enhance fiscal capacity by boosting economic growth.

The National Health Policy aims to achieve the following goals:

- It aims to offer superior health services to every age group and gender.
- The policy focuses on providing universal access to excellent quality health care services at a reasonable cost.
- Promoting health care orientation in every developmental policy.
- Offering access to better treatment, lowering expenses related to health care services and improving quality.
- It aims to reduce premature mortality from cancer, cardiovascular diseases, chronic respiratory diseases and diabetes by 25% within 2025.
- This policy recognises the importance of sustainable development and time-bound quantitative goals.
- National Health Policy in India improves overall health status through promotive, palliative, and rehabilitative services

Through Various health schemes in india

- Aam Aadmi Bima Yojana.
- Central Government Health Scheme.
- Ayushman Bharat Yojana.
- Universal Health Insurance Scheme.
- Rashtriya Swasthya Bima Yojana.
- Mahatma Jyotiba Phule Jan Arogya Yojana.
- West Bengal Health Scheme.
- Niramaya Health Insurance Scheme.

Research Ethics

Research ethics involves the application of fundamental ethical principles to research activities which include the design and implementation of research, respect towards society and others, the use of resources and research outputs, scientific misconduct and the regulation of research.

7 principles of ethics in research

NIH Clinical Center researchers published seven main principles to guide the conduct of ethical research:

- Social and clinical value.
- Scientific validity.
- Fair subject selection.
- Favorable risk-benefit ratio.
- Independent review.
- Informed consent.
- Respect for potential and enrolled subjects.

Honesty: Strive for honesty in all scientific communications. Honestly report data, results, methods and procedures, and publication status. Do not fabricate, falsify, or misrepresent data. Do not deceive colleagues, research sponsors, or the public.

Integrity: Keep your promises and agreements; act with sincerity; strive for consistency of thought and action.

Carefulness: Avoid careless errors and negligence; carefully and critically examine your own work and the work of your peers. Keep good records of research activities, such as data collection, research design, and correspondence with agencies or journals.

Openness: Share data, results, ideas, tools, resources. Be open to criticism and new ideas.

Transparency: Disclose methods, materials, assumptions, analyses, and other information needed to evaluate your research.

Accountability: Take responsibility for your part in research and be prepared to give an account (i.e. an explanation or justification) of what you did on a research project and why.

Intellectual Property: Honor patents, copyrights, and other forms of intellectual property. Do not use unpublished data, methods, or results without permission. Give proper acknowledgement or credit for all contributions to research. Never plagiarize.

Confidentiality: Protect confidential communications, such as papers or grants submitted for publication, personnel records, trade or military secrets, and patient records.

Responsible Publication: Publish in order to advance research and scholarship, not to advance just your own career. Avoid wasteful and duplicative publication.

Responsible Mentoring: Help to educate, mentor, and advise students. Promote their welfare and allow them to make their own decisions.

Respect for Colleagues: Respect your colleagues and treat them fairly.

Social Responsibility: Strive to promote social good and prevent or mitigate social harms through research, public education, and advocacy.

Non-Discrimination: Avoid discrimination against colleagues or students on the basis of sex, race, ethnicity, or other factors not related to scientific competence and integrity.

Competence: Maintain and improve your own professional competence and expertise through lifelong education and learning; take steps to promote competence in science as a whole.

Legality: Know and obey relevant laws and institutional and governmental policies.

Animal Care: Show proper respect and care for animals when using them in research. Do not conduct unnecessary or poorly designed animal experiments.

Animal Research Ethics

prepared by the National Committee for Research Ethics in Science and Technology (NENT). Their purpose is to provide ethical guidelines for researchers and other people who are considering experiments on animals. The guidelines will be useful when planning projects, assessing them, and when reporting and publishing findings and results. They are also intended to contribute to reflection on research ethics and the use of animals in research in both research communities and in the public debate.

Ethics and Experiments on Animals

It is generally thought that it may be necessary to use laboratory animals in some cases in order to create improvements for people, animals or the environment. At the same time, the general opinion is that animals have a moral status, and that our treatment of them should be subject to ethical considerations. Such views are reflected in the following positions:

- Animals have an intrinsic value which must be respected.
- Animals are sentient creatures with the capacity to feel pain, and the interests of animals must therefore be taken into consideration.
- Our treatment of animals, including the use of animals in research, is an expression of our attitudes and influences us as moral actors.

The ethical **guidelines** provided encompass a comprehensive framework for the responsible use of laboratory animals in research. Here's a synthesized version:

- **Respect for Animals' Dignity:** Researchers must recognize animals' inherent worth and interests, ensuring respectful treatment and tailored care.
- **Responsibility for Considering Options (Replace):** Researchers should prioritize

alternatives to animal experimentation whenever feasible, considering postponement if suitable alternatives are unavailable.

- **Principle of Proportionality:** Researchers must weigh potential benefits against animal suffering, ensuring that any harm is justified by substantial and probable benefits to animals, humans, or the environment.
- **Responsibility for Reducing Animal Numbers (Reduce):** Researchers should minimize the number of animals used, employing literature studies and alternative designs to maintain scientific quality.
- **Responsibility for Minimizing Suffering (Refine):** Researchers must minimize suffering before, during, and after experiments, addressing all aspects of animal welfare and considering adaptation periods.
- **Responsibility for Maintaining Biological Diversity:** Researchers must prevent harm to ecosystems and endangered species, observing the precautionary principle when uncertain outcomes are possible.
- **Responsibility for Openness and Sharing:** Researchers must ensure transparency in research findings, sharing data and materials to avoid redundant experiments and enhance public awareness.
- **Requirement of Due Care:** Researchers and managers must comply with national and international regulations governing the use of laboratory animals, familiarizing themselves with current laws.

Quiz:

1. What is the role of antibiotics in poultry?
2. What are allergens in food? Give an example.
   Ex: cow's milk, peanuts, eggs, fish
3. What is Research ethics?
   Research ethics (research integrity) is a set of guidelines and norms that guide researchers in conducting their studies with integrity, respect and adherence to human rights.These principles help to ensure that research is conducted properly and conscientiously and that the findings and theories are accurately reported and discussed.
4.